STEP-BY-STEP IMAGE PRE-PROCESSING IN FIJI – NFH DEGENERATION

Timing: ~15-20 min for parameter selection; ~10-30s per file for automated processing

The image pre-processing steps are essential for the accurate segmentation of axons (NFH) and the removal of noise prior to object identification. The Fiji macros described below perform batch processing to determine threshold parameters, generate binary masks, and prepare images for subsequent CellProfiler analysis.

All scripts and pipelines can be found at the GitHub repository

Stage 1: Image Pre-processing in Fiji/ImageJ

1. Open Fiji and run the threshold selection macro ("Choose threshold.ijm") on the input image folder. This macro opens a random subset of images to allow the user to determine optimal thresholding parameters.

   a. Select Plugins > Macros > Run....

   b. Select the macro "Choose threshold.ijm" in the popup window and click Open.

   c. In the prompt window, input the channel number corresponding to NFH (e.g., enter 3 if NFH is the third channel).

   d. Select the input folder containing the raw .ims or .tif images.

   e. Enter the number of random images to open for testing (the maximum available count will be displayed).

   f. Wait until the selected images are opened, channels are split, and the channel of interest is displayed.

2. Determine the optimal threshold method and range.

   a. Select Image > Adjust > Threshold... (or press Shift + Control + T).

   b. Adjust the threshold method and range to capture the maximum amount of NFH signal while minimizing noise.

   c. CRITICAL: Ensure that the chosen parameters distinguish degenerated NFH from background noise; if intensities are identical or , segmentation may fail.

   d. Record the Threshold Method, Threshold Minimum, and Threshold Maximum values.

e. Close all images (Press Shift + W).

Note: This macro is intended for visual inspection and parameter optimization only; no output files are generated. Use this step to identify and record the threshold method, minimum, and maximum values required for the subsequent processing macro.

3. Run the processing macro ("Process Threshold.ijm") to generate masks. This macro applies the parameters determined in Step 2 to the entire dataset.

   a. Select Plugins > Macros > Run....

   b. Select the macro "Process Threshold.ijm" and click Open.

   c. In the prompt window, input the name of the channel to be used as a mask (e.g., NFH).

   d. Enter the parameters recorded in Step:

   i. Channel Number.

   ii. Threshold Method, Min, and Max.

   e. Enter the Mean Cutoff value (default is 0.95).

   f. Click OK to run the script.

   g. This will output a folder named "Channel_X_MASKNAME" (e.g., "Channel_1_NFH") and a "pixel metadata" file containing pixel dimensions (μm²).

   Note: The **Mean Cutoff** parameter is a quality control step. It calculates the mean intensity (0 to 1) for each slice. Slices with a mean intensity higher than the set value (e.g., 0.95) are automatically deleted to remove artifacts where thresholding is calculated incorrectly. The default value is typically sufficient to ensure the removal of artifact slices.

Analysis using CellProfiler

Timing: ~5-10 min to set optimal parameters; ~10-20s to analyze each file

This section outlines the analysis pipeline for the identification and quantification of intact and degenerated NFH. The pipeline utilizes distinct modules for channel separation, object segmentation, and morphological measurement.

1. Initialize the CellProfiler project and load the pipeline.

    a. Open CellProfiler.

    b. Import the pipeline file “NFHdeg.cppipe” by dragging and dropping it into the pipeline panel.

    c. Select the "Images" tab and drag and drop the pre-processed image folders (from the previous section) into the file list.

2. Configure output image

    a. Select "Output settings" to define the destination directory for the output images and for the generated .csv spreadsheets. It is recommended to create a new folder where the output will be stored.

3. To the left of each module, there is the icon of an eye (or). This indicates which modules will be visible while running the pipeline. During the test mode, we recommend making the following modules visible: ColorToGray, IdentifyPrimaryObjects (both instances), OverlayOutlines.

4. Optimize segmentation parameters in Test Mode. Select "Start Test Mode" to begin stepwise optimization.

<u>Table 1: Cell Profiler modules</u>

Module	Function	Action Required
Images	Select images to undergo analysis	Upload images from the previous preprocessing step
Metadata	Option to extract metadata	Not necessary to extract metadata
NamesAndTypes	Identify and name images	Certify that the assigned name is “NFHimage_”
Groups	Option to group images	Not necessary to extract metadata
ColorToGray	Converts color images to grayscale	Make sure “Channel number” is set to “1” and “Image name” is set to “NFH”
IdentifyPrimaryObjects	Identifying objects from a binary image for subsequent calculation	Adjust “Typical diameter of objects” and other parameters until **intact** NFH is selected

ConvertObjectsToImage	Convert set of identified intact NFH objects to static image	No action needed
SaveImages	Saving the previously created image to output path	Change file name under "Enter file prefix" if desired
IdentifyPrimaryObjects	Identifying objects from a binary image for subsequent calculation	Adjust "Typical diameter of objects" and other parameters until **degenerated** NFH is selected
ConvertObjectsToImage	Converting set of identified degenerated NFH objects to static image	No action needed
SaveImages	Saving the previously created image to output path	Change file name under "Enter file prefix" if desired
OverlayOutlines	Placing identified objects over a binary image for ease of visualization	Change colors as desired
SaveImages	Saving the previously created image to output path	Change file name under "Enter file prefix" if desired
MeasureImageAreaOccupied	Calculating area of identified objects	No action needed
MeasureObjectSizeShape	Calculating number of identified objects	No action needed
ExportToSpreadsheet	Exporting results to .csv files in the output path	No action needed.

2. Run the analysis.

    a. Once settings are optimized in Test Mode, select "Exit Test Mode".

    b. Select "Analyze Images" to execute the pipeline on the full dataset.

    c. Upon completion, the data will be exported as .csv files (e.g., "Results.csv") in the designated output directory.

Note: i) For statistical analysis, the "AreaOccupied" and "Count" columns in the output spreadsheet should be used. To convert pixel area to μm², multiply the pixel count by the pixel area value located in the "pixel metadata" file generated during the preprocessing stage. Objects that are not segmented properly can be identified by

reviewing the output overlay images and should be excluded from downstream analysis.
ii) The “Action required” tab for the IdentifyPrimaryObjects module is a suggestion. The user should know that there are a number of parameters which can be modified and it is the user’s responsibility to make sure those parameters are adjusted properly